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Introduction and Overview

This manual guides coaches through a process to effectively deliver an integrated training program to golfers. As mentioned throughout this course, the Optimum Performance Training (OPT) model is analogous to a staircase, guiding clients through different levels of physical, physiological, and psychological adaptation intended to help achieve specific goals and outcomes. This process will involve going up and down the stairs, stopping at different steps, and moving to various heights, depending on the client's goals, needs, and abilities. It is important to note that the coach may need to revisit phases throughout training to ensure that their clients maintain proper movement patterns and safety.

The manual will take you through general goals and conditioning elements that coaches must consider when working with golfers. It will also identify appropriate assessments so the coach can design individualized exercise programs based on the client's goals, abilities, comfort level, and movement impairments. The manual will also provide progressive yearly and monthly plans to implement a global conditioning program safely and effectively. This manual includes sample daily integrated training programs utilizing NASM's corrective exercise process and phases 1, 2, 3, 5, and 6 of the OPT Model. While the client may be able to perform all phases of OPT successfully, these four phases are the most commonly used when working with golfers. The example exercise programs will guide the coach on enhancing overall health while focusing on their client's quality of movement, stabilization, strength, and power.

We hope you find this manual helpful when working with golfers!

Goals and Considerations

GOLFERS CONDITIONING GOALS AND CONSIDERATIONS

- Correct muscle imbalances that may predispose clients to injury.
- Improve flexibility in all three planes of motion.
- Improve total-body control that can carry over to functional movement and activities on the golf course.
- Improve joint motion to facilitate proper alignment and movement.
- Improve core stabilization strength and endurance to maintain stability for extended periods.
- Incorporate corrective exercise to improve neuromuscular function and decrease injury risk.
- Integrate core, balance, and plyometric training into the program to improve proprioceptive skills and neuromuscular control.
- Increase stabilization, strength, and power based on the client's goals and assessment results to improve golfing ability.

OPT PROGRAMMING GOALS AND CONSIDERATIONS

- To provide continued enhancements to movement patterns and muscle imbalances.
- Enhance range of motion, joint stability, strength, and power to maintain optimal functional performance and decrease the risk of injury.
- Maintain cardiorespiratory and muscular endurance to facilitate better golf performance.
- Consider the exercise volume and training phases used in the plan based on the client's current fitness level and goals.
- Incorporate the power phase into the program to improve the rate of force production for enhanced clubhead speed and driving distance.
- Promote exercises with multiplanar movements to improve strength and coordination in all three planes of motion.
- Modify exercise positions based on the client's capabilities and conditions.

Golfer Assessment Process

The coach should perform a battery of assessments to identify their client's readiness for exercise and current fitness status. The following provides a list of example assessments that apply to a majority of golfing clients. However, this is not an exhaustive list. It is up to the coach's discretion and expertise to determine which assessments are most appropriate for their client.

Example Subjective Assessments

- To determine client goals, lifestyle, and medical history
 - PAR-Q+
 - General lifestyle and medical history questionnaire

Example Physiological and Anthropometric Assessments

- To determine current physiologic and anthropometric baseline measurements
 - Resting heart rate
 - Blood pressure
 - Circumference measurements
 - Scale weight and BMI

Example Cardiorespiratory Assessments

- To determine aerobic fitness
 - Rockport Walk Test or YMCA 3-minute Step Test
 - 1.5 Mile Run Test

Example Dynamic Postural Assessments

- To determine potential muscle imbalances, range of motion, and movement impairments

- Overhead Squat Assessment
- Pushing / Pulling Assessments

Example Mobility Assessments

- To determine the range of motion of individual joint segments
 - Weight-bearing lunge Test
 - Modified Thomas Test
 - Hip Internal/External Rotation tests
 - Modified Thomas Test
 - Shoulder Flexion, Internal/External Rotation tests

Example Strength Assessments

- To determine various muscle force capabilities
 - Grip Strength Assessment
 - 1-5 RM Squat Test
 - 1-5 RM Bench Press Test

Example Power Assessments

- To determine the rate of force production of various movement patterns and muscle groups
 - Vertical Jump Test
 - Rotational Medicine Ball Throw Test
 - Seated Medicine Ball Chest Pass Test

Example Annual Program

ANNUAL PLAN													
LEVEL	MONTH	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
	PHASE												
Corrective	CES	√	√	√	√	√	√	√	√	√	√	√	√
Stabilization	1	√	√			√	√	√	√	√	√	√	√
Strength	2			√		√	√	√	√	√	√	√	√
	3				√								
	4												
Power	5					√	√						
	6							√	√	√	√	√	√

Above is a sample annual training plan based on a golfer's general health and fitness goal. Adjustments may be made based on the client's fitness level and functional ability. The annual program will improve stability, balance, coordination, and enhance strength and power to assist with golfing ability. Cardiorespiratory training is also implemented 2-5 days per week for 30-60 minutes, depending on the client's fitness level, goals, schedule, and assessment results. It is important to note that some clients may need to work up to 30 minutes of cardiorespiratory training, so the coach should program accordingly to fit each client's needs, abilities, and time commitment.

Corrective exercise is included in multiple months up to 5 days a week. This inclusion aims to focus on movement impairments that may have been noted in the assessment evaluation process and to ensure proper movement patterns are maintained throughout their golf fitness journey. The client should also be encouraged to perform these exercises independently as *homework*. Corrective exercise programs can also be implemented as a "warm-up" routine before the scheduled workout. Therefore, the annual plan includes corrective exercise in all twelve months.

Example Monthly/Weekly Programs

MONTHLY/WEEKLY PLAN (Months 1 and 2)																												
WEEK	1							2							3							4						
DAY	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S
PHASE																												
CES		√		√		√			√		√		√			√		√		√			√		√		√	
1	√		√		√			√		√		√			√		√		√			√		√		√		
2																												
3																												
4																												
5																												
Cardio		√		√					√		√					√		√					√		√			
Re-assessment																											√	

Months 1 and 2: Focus on correcting muscle imbalances and improving stability. Starting in month 2, the coach can increase the intensity and demand of each exercise routine if deemed appropriate for the client, and proper movement patterns are maintained throughout the program. Following this period, Phase 2 can be introduced to enhance stability while increasing prime mover strength if the assessment results show the client is ready to advance.

Corrective exercise can be included as part of the client's warm-up and performed as homework on the weekends and before cardio.

MONTHLY/WEEKLY PLAN (Month 3)																												
WEEK	1							2							3							4						
DAY	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S
PHASE																												
CES			√			√				√			√				√			√				√			√	
1																												
2	√	√		√	√			√	√		√	√			√	√		√	√			√	√		√	√		
3																												
4																												
5																												
Cardio			√			√				√			√				√			√				√			√	

Month 3: Focus on progressing stabilization exercises while enhancing prime mover strength by incorporating supersets, as seen in Phase 2.

These supersets may need to be adjusted to fit the client's needs and abilities. The sample monthly program shows the client performing Phase 2 workouts four days per week. If this fits the client's schedule, upper and lower body split routines can be implemented to switch up their previous routine and decrease the chances of overtraining. Cardio and corrective exercises can be included on days off from Phase 2 workouts to maintain proper muscle activation, movement patterns, and joint stability and mobility.

MONTHLY/WEEKLY PLAN (Month 4)																												
WEEK	1							2							3							4						
DAY	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S
PHASE																												
CES			√			√				√			√				√			√				√			√	
1																												
2																												
3	√	√		√	√			√	√		√	√			√	√		√	√			√	√		√	√		
4																												
5																												
Cardio			√			√				√			√				√			√				√			√	
Re-assessment																										√		

Month 4: Focus is primarily on strength and hypertrophy, so there will be an increase in training volume and intensity performed throughout this mesocycle. Strength and hypertrophy will assist the golfer in developing the attributes needed for power development later in the client's program. Like the previous month, the sample monthly program shows the client exercising in Phase 3 four days per week. If this fits the client's schedule, upper and lower body split routines or targeting specific body parts can be implemented to offer a new type of routine.

MONTHLY/WEEKLY PLAN (Months 5 and 6)																												
WEEK	1							2							3							4						
DAY	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S
PHASE																												
CES		✓		✓		✓			✓		✓		✓			✓		✓		✓			✓		✓		✓	
1			✓							✓							✓							✓				
2	✓							✓							✓							✓						
3																												
4																												
5					✓							✓							✓							✓		
Cardio		✓		✓					✓		✓					✓		✓					✓		✓			
Re-assessment																									✓			

Months 5 and 6: This month uses undulated periodization, aiming for simultaneous stabilization, strength, and power adaptations. However, training for speed and power should best fit the golfing client's needs and abilities because it is an advanced form of conditioning. Power training is beneficial, but this inclusion should be based on previous experiences and the success of exercise execution throughout the OPT model, along with how the client moves and their desired goals.

It is essential to include corrective exercises due to the higher intensities associated with a power training routine. It is also strongly suggested that Phase 1 and 2 programs be integrated throughout the week to help maintain proper movement quality and decrease the risk of overtraining.

MONTHLY/WEEKLY PLAN (Months 7 through 12)																												
WEEK	1							2							3							4						
DAY	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S
PHASE																												
CES		✓		✓		✓			✓		✓		✓			✓		✓		✓			✓		✓		✓	
1			✓							✓							✓							✓				
2	✓							✓							✓							✓						
3																												
4																												
5																												
6					✓							✓							✓							✓		
Cardio		✓		✓					✓		✓					✓		✓					✓		✓			
Re-assessment																												

Months 7 through 12: Months 7 through 12 undulate between Phases 1, 2, and 6. These months' goals are to continue increasing stabilization, strength, and power adaptations to aid in golf performance. Phase 6. Maximal Power Training focuses on high-velocity training to further develop the biomotor abilities of power and speed necessary to increase clubhead speed and driving distance. The client must approach each exercise with adequate intensity, focusing on accelerating through an exercise's entire range of motion.

However, if deemed necessary and appropriate, the coach can add or substitute other phases through this mesocycle, such as Phase 3 or Phase 5.

Example Corrective Exercise Programs

Corrective exercise is a term used to describe the systematic process of identifying a neuromusculoskeletal dysfunction, developing a plan of action, and implementing an integrated corrective strategy. Each corrective exercise plan includes four steps. The first step is the Inhibit phase using inhibitory techniques. Inhibitory techniques are used to reduce tension or decrease the activity of overactive neuromyofascial tissues in the body. This can be accomplished with self-myofascial techniques (e.g., foam rolling). The second step is the Lengthen phase, using lengthening techniques. Lengthening techniques increase the extensibility and range of motion (ROM) of neuromyofascial tissues in the body. This can be accomplished using static stretching and neuromuscular stretching. The third step is the Activate phase using activation techniques. Activation techniques are used to reeducate or increase the activation of underactive tissues as identified during the assessment process. This can be accomplished using isolated (single-joint) strengthening exercises. The fourth and final step is the Integrate phase using integration techniques. Integration techniques are used to retrain the collective synergistic function of all muscles through functionally progressive exercise using integrated dynamic movements (multi-joint, compound exercises).

Below are examples of corrective exercise programs based on common movement compensations the coach may encounter. These programs can be used as a standalone or integrated into an OPT workout as a specialized warm-up. As with all programs, modifications may need to be made to ensure client safety and proper execution. If compensations start to occur, it is the coach's responsibility to regress the exercise or select a different exercise so the client learns proper movement patterns that will carry over into activities of daily living and golf.



Foot and Ankle Dysfunction (feet turn out)

EXERCISE	SETS	REPS	TEMPO	REST	NOTES
INHIBIT (self-myofascial techniques)					
Bilateral Calf Complex	1	1	Slow	0	Hold tender areas for 30 seconds
Bilateral Outer Hamstring (Biceps Femoris -short head)	1	1	Slow	0	Hold tender areas for 30 seconds
Bilateral Tensor Fascia Latae (TFL)	1	1	Slow	0	Hold tender areas for 30 seconds
LENGTHEN (static stretching)					
Bilateral Calf Stretch	1	1	Slow	0	Hold each stretch for 30 seconds
Bilateral Outer Hamstring Stretch	1	1	Slow	0	Hold each stretch for 30 seconds
Bilateral TFL Stretch	1	1	Slow	0	Hold each stretch for 30 seconds
ACTIVATION (isolated strengthening techniques - performed in a circuit)					
Calf Raises -Toes Turned Slightly Inward	2	10	Slow	0	Maintain proper alignment
Ball Hamstring Curls -Toes Turned Inward	2	10	Slow	0	Maintain neutral spine
Floor Bridge	2	10	Slow	60 sec	Maintain a neutral spine and squeeze glutes at the top
INTEGRATION (multi-joint, compound exercises)					
Single-Leg Balance with Reach: Frontal Plane	2	10	Slow	60 sec	Maintain proper arch alignment

Knee Dysfunction (knee's cave inward)

EXERCISE	SETS	REPS	TEMPO	REST	NOTES
INHIBIT (self-myofascial techniques)					
Bilateral Calf Complex	1	1	Slow	0	Hold tender areas for 30 seconds
Bilateral Adductor Complex	1	1	Slow	0	Hold tender areas for 30 seconds
Bilateral Outer Hamstring (Biceps Femoris -short head)	1	1	Slow	0	Hold tender areas for 30 seconds
Bilateral TFL	1	1	Slow	0	Hold tender areas for 30 seconds
LENGTHEN (static stretching)					
Bilateral Calf Stretch	1	1	Slow	0	Hold each stretch for 30 seconds
Standing Adductor Stretch	1	1	Slow	0	Hold each stretch for 30 seconds
Bilateral Outer Hamstring Stretch	1	1	Slow	0	Hold each stretch for 30 seconds
Bilateral TFL Stretch	1	1	Slow	0	Hold each stretch for 30 seconds
ACTIVATION (isolated strengthening techniques - performed in a circuit)					
Side-Lying Leg Raise	2	10	Slow	0	Maintain proper alignment
Cable Terminal Knee Extensions (TKE's)	2	10	Slow	0	Fully extend the knee and squeeze glutes
Standing Cable Leg Curl with Toes Turned Inward	2	10	Slow	60 sec	Maintain neutral spine
INTEGRATION (multi-joint, compound exercises - performed in a circuit)					
Lateral Tube Walks	2	10	Slow	0	Keep toes and knees pointed forward
Ball Squats	2	10	Slow	60 sec	Be sure to squeeze the glutes at the top

Lumbo-Pelvic-Hip Dysfunction (forward lean and anterior pelvic tilt)

EXERCISE	SETS	REPS	TEMPO	REST	NOTES
INHIBIT (self-myofascial techniques)					
Bilateral Calf Complex	1	1	Slow	0	Hold tender areas for 30 seconds
Bilateral TFL	1	1	Slow	0	Hold tender areas for 30 seconds
Bilateral Lats	1	1	Slow	0	Hold tender areas for 30 seconds
LENGTHEN (static stretching)					
Bilateral Calf Stretch	1	1	Slow	0	Hold each stretch for 30 seconds
Bilateral Standing Hip Flexor Stretch	1	1	Slow	0	Hold each stretch for 30 seconds
Bilateral Ball Lat Stretch	1	1	Slow	0	Hold each stretch for 30 seconds
ACTIVATION (isolated strengthening techniques - performed in a circuit)					
Bird Dog	2	10	Slow	0	Maintain proper alignment
Plank	2	10	Slow	0	Maintain a neutral spine.
Band Resisted Clamshells	2	10	Slow	60 sec	Maintain a neutral spine. Use light band
INTEGRATION (multi-joint, compound exercises - performed in a circuit)					
Cable Squat to Row	2	10	Slow	0 sec	Maintain proper alignment
Side Step-Up to Balance (frontal plane)	2	10	Slow	60 sec	Squeeze glutes at the top

Shoulder Dysfunction (arms fall forward)

EXERCISE	SETS	REPS	TEMPO	REST	NOTES
INHIBIT (self-myofascial techniques)					
Bilateral Pecs	1	1	Slow	0	Hold tender areas for 30 seconds
Bilateral Lats	1	1	Slow	0	Hold tender areas for 30 seconds
Thoracic Spine	1	1	Slow	0	Hold tender areas for 30 seconds
LENGTHEN (static stretching)					
Bilateral Wall Pec Stretch	1	1	Slow	0	Hold each stretch for 30 seconds
Bilateral Ball Lat Stretch	1	1	Slow	0	Hold each stretch for 30 seconds
Bilateral Sleeper Stretch	1	1	Slow	0	Hold each stretch for 30 seconds
ACTIVATION (isolated strengthening techniques - performed in a circuit)					
Floor Cobra	2	10	Slow	0	Externally rotate arms
Wall Angels	2	10	Slow	0	Maintain a neutral spine
Banded Face Pulls	2	10	Slow	60 sec	Maintain a neutral spine. Use light band
INTEGRATION (multi-joint, compound exercises - performed in a circuit)					
Cable Squat to Row	2	10	Slow	0 sec	Maintain proper alignment
Ball Combo I	2	10	Slow	60 sec	YTA's on a stability ball

Example Stabilization Endurance Training Programs

The first level of training in the OPT model focuses on two main adaptations: (1) developing proper movement patterns and (2) the mobility and stability of the entire kinetic chain. This level of the OPT model is also designed to prepare the body for the demands of higher levels of training that follow. This period is crucial for all beginners to help instill confidence and promote adherence to an exercise program while minimizing stress experienced at higher intensities of training that could ultimately lead to burnout or injury. It is also necessary to cycle back through this level after periods focused on strength and power training to maintain a high degree of core and joint stability. In addition, this level allows the body actively to rest and recover from more intense bouts of training. The stabilization level focuses on the following:



- Developing proper movement patterns: squat, push, pull, press, hip hinge, and multiplanar movement
- Acquiring basic exercise skills and proper use of exercise machines, free weights, and cardio equipment
- Correcting muscle imbalances
- Improving activation of the core musculature and stabilization of the spine and pelvis
- Enhancing balance, coordination, and postural control during physical activity, exercise, and sport
- Preventing tissue overload by preparing muscles, tendons, ligaments, and joints for the upcoming imposed demands of training
- Improving cardiorespiratory and muscular endurance
- Promoting client confidence and adherence to exercise
- The stabilization level of training in the OPT model consists of one phase: Phase 1. Stabilization Endurance Training.

Phase 1. Stabilization Endurance Training should be viewed as a phase of training intended to build a stable foundation so the client can move, feel, and perform at their best. Because there are many similarities and complementary variables, the components of corrective exercise training can be blended and incorporated nicely into a Phase 1 training program if the coach feels this would benefit the client.

Phase 1. Stabilization Endurance Training (program 1)

EXERCISE	SETS	REPS	TEMPO	REST	NOTES
WARM-UP					
SMT: Bilateral Calves, Adductors, Lats	1	1	Slow	0	Hold tender areas for 30 seconds
Static Stretch: Bilateral Calves, Adductors, and Lats	1	1	Slow	0	Hold each stretch for 30 seconds
ACTIVATION (core & balance - performed in a circuit)					
Floor Bridge	1	15	Slow	0	Keep glutes and abs engaged
Pallof Press	1	15	Slow	0	Keep glutes and abs engaged
Multiplanar Single-leg Balance Reach	1	6 each	Slow	60 sec	Keep glutes and abs engaged
SKILL DEVELOPMENT (plyometric & SAQ)					
Squat Jump with Stabilization	1	5	Controlled	60 sec	Hold landing for 3-5 seconds each
RESISTANCE TRAINING (performed in a circuit)					
Cable Squat to Row	1	15	Slow	0	Maintain the 5 kinetic chain checkpoints
Push-up	1	15	Slow	0	Can perform on wall or bench if needed
Stability Ball Cobra	1	15	Slow	0	Triple Extension
Single-Leg Scaption	1	15	Slow	0	Maintain proper alignment
Side-Lying Leg Raises	1	15 each	Slow	60 sec	Be sure to keep torso straight
CLIENT'S CHOICE					
					Insert client's chosen exercise
COOL-DOWN					
SMT: Bilateral Calves, Adductors, Lats	1	1	Slow	0	Hold tender areas for 30 seconds
Static Stretch: Bilateral Calves, Adductors, and Lats	1	1	Slow	0	Hold each stretch for 30 seconds

Phase 1. Stabilization Endurance Training (program 2)

EXERCISE	SETS	REPS	TEMPO	REST	NOTES
WARM-UP					
SMT: Bilateral Calves, Piriformis, Lats	1	1	Slow	0	Hold tender areas for 30 seconds
Static Stretch: Bilateral Calves, Piriformis, and Lats	1	1	Slow	0	Hold each stretch for 30 seconds
ACTIVATION (core & balance - performed in a circuit)					
Floor Glute Bridges	2	12	Slow	0	Keep glutes and abs engaged
Planks	2	1	Slow	0	Hold plank for 30 sec
Single-Leg Lift and Chop	2	6 each	Slow	60 sec	Use a light medicine ball
SKILL DEVELOPMENT (plyometric & SAQ)					
Lateral Squat Jump with Stabilization (frontal plane)	2	5	Controlled	0	Hold landing for 3-5 seconds each
RESISTANCE TRAINING (performed in a circuit)					
Ball Squat to Overhead Press	2	12	Slow	0	Maintain the 5 kinetic chain checkpoints
Staggered Stance Cable Chest Press	2	12	Slow	0	Include triple extension on back stance leg
Staggered Stance Cable Rows	2	12	Slow	0	Be sure not to shrug shoulders
Ball Hamstring Curls	2	12	Slow	60 sec	Maintain proper alignment
CLIENT'S CHOICE					
					Insert client's chosen exercise
COOL-DOWN					
SMT: Bilateral Calves, Piriformis, Lats	1	1	Slow	0	Hold tender areas for 30 seconds
Static Stretch: Bilateral Calves, Piriformis, and Lats	1	1	Slow	0	Hold each stretch for 30 seconds

Phase 1. Stabilization Endurance Training (program 3)

EXERCISE	SETS	REPS	TEMPO	REST	NOTES
WARM-UP					
SMT: Bilateral Calves, Piriformis, Thoracic Spine	1	1	Slow	0	Hold tender areas for 30 seconds
Static Stretch: Bilateral Calves, Hip Flexors, and Lats	1	1	Slow	0	Hold each stretch for 30 seconds
ACTIVATION (core & balance - performed in a circuit)					
Ball Glute Bridges	2	12	4/2/1/0	0	Maintain neutral spine
Side Planks	2	1	Controlled	0	Hold for 30 sec
Single-Leg Throw and Catch	2	6 each	Controlled	60 sec	Use a light medicine ball
SKILL DEVELOPMENT (plyometric & SAQ)					
Turning Squat Jump with Stabilization (transverse plane)	2	5	Controlled	60 sec	Hold landing for 3-5 seconds each
RESISTANCE TRAINING (performed in a circuit)					
Cable Squat to Row	3	12	Slow	0	Progress to single-leg stance if appropriate
Suspended Push-up	3	12	Slow	0	Use a suspension trainer
Ball Cobra	3	12	Slow	0	Retract and depress shoulders
Single-leg Romanian Deadlift	3	12	Slow	60 sec	Keep a neutral spine
CLIENT'S CHOICE					
					Insert client's chosen exercise
COOL-DOWN					
SMT: Bilateral Calves, Piriformis, Thoracic Spine	1	1	Slow	0	Hold tender areas for 30 seconds
Static Stretch: Bilateral Calves, Hip Flexors, and Pecs	1	1	Slow	0	Hold each stretch for 30 seconds

Example Strength Training Programs

The second level of training in the OPT model focuses on the main adaptation of strength. While there are many types of strength, the OPT model emphasizes strength endurance, muscular development, and maximal strength improvements. It is designed to maintain stability while increasing the amount of stress placed on the body for increased muscle size, strength, and work capacity. This period of training is a necessary progression from stabilization for golfers who desire to increase caloric expenditure, muscle size, muscle strength, and bone mineral density.

The strength level of training focuses on the following:

- Increasing the ability of the core musculature to stabilize the pelvis and spine under heavier loads through greater ranges of motion
- Increasing the load-bearing capabilities of muscles, tendons, ligaments, and joints
- Increasing the volume of training
- Increasing metabolic demand by taxing the phosphocreatine (ATP-PC) and glycolytic energy systems to induce cellular changes in muscle
- Increasing recruitment of more motor units to overcome an external load (maximal strength)

The strength level of training in the OPT model consists of three phases: Phase 2. Strength Endurance Training, Phase 3. Muscular Development Training, and Phase 4. Maximal Strength Training.

Phase 2. Strength Endurance Training is a hybrid form of training that promotes increased stabilization endurance, hypertrophy, and strength. The unique aspect of this phase of training is that it entails the use of superset techniques in which a more strength-focused exercise (such as a bench press) is immediately followed by a stabilization-focused exercise with similar biomechanical motions (such as a push-up). Thus, for every set performed, two exercises are completed. High volumes can be generated during this phase of training.

Phase 3 Muscular Development Training is specific for the adaptation of maximal muscle growth, focusing on high levels of volume to force cellular changes that result in an overall increase in muscle size. Because the goal of this phase of training is primarily hypertrophy, the coach will

want to increase the volume and intensity of the program. A client in this category will generally stay in this phase of training for a 2- to 6-week duration before cycling back through Phase 1 or 2 or progressing on to Phases 4, 5, or 6.

Phase 4. Maximal Strength Training focuses on increasing the load placed on the tissues of the body. In other words, Phase 4 requires the exerciser to lift progressively heavier loads near or at maximal intensity. Maximal intensity improves the recruitment of more motor units and the rate of force production.

Depending on the client's goals, most often, Phases 2 and 3 are most appropriate for golfers, so examples of these phases are provided.

Phase 2. Strength Endurance Training (program 1)

EXERCISE	SETS	REPS	TEMPO	REST	NOTES
WARM-UP					
SMT: Bilateral Calves, TFL, Lats	1	1	Slow	0	Hold tender areas for 30 seconds
Active Stretch: Bilateral Calves, Hip Flexors, and Lats	1	8	Medium	0	Hold each stretch for 1-2 seconds
ACTIVATION (core & balance - performed in a circuit)					
Ball Crunch with Rotation	2	10	Medium	0	Keep abs engaged
Single-Leg Squat	2	8 each	Medium	60 sec	Maintain knee alignment
SKILL DEVELOPMENT (plyometric & SAQ)					
RESISTANCE TRAINING (vertical loading protocol)					
Chest Press Machine	2	10	Medium	0	Superset
Push-ups		10	Slow	60 sec	
Bent-over Barbell Rows	2	10	Medium	0	Superset
Stability Ball Dumbbell Cobras		10	Slow	60 sec	
Alternating Dumbbell Lunge	2	10 each	Medium	0	Superset
Dumbbell Step-Up to Balance		10 each	Slow	60 sec	
CLIENT'S CHOICE					
					Insert client's chosen exercise
COOL-DOWN					
SMT: Bilateral Calves, TFL, Lats	1	1	Slow	0	Hold tender areas for 30 seconds
Static Stretch: Bilateral Calves, Hip Flexors, and Lats	1	1	Slow	0	Hold each stretch for 30 seconds

Phase 2. Strength Endurance Training (program 2)

EXERCISE	SETS	REPS	TEMPO	REST	NOTES
WARM-UP					
SMT: Bilateral Calves, Piriformis, Thoracic Spine	1	1	Slow	0	Hold tender areas for 30 seconds
Active Stretch: Bilateral Calves, Hip Flexors, and Pecs	1	8	Medium	0	Hold each stretch for 1-2 seconds
ACTIVATION (core & balance)					
Cable Rotations	2	8	Medium	0	Triple extension of rear leg
Side Lunge to Balance (frontal plane)	2	8 each	Medium	60 sec	Maintain knee alignment
SKILL DEVELOPMENT (plyometric & SAQ)					
RESISTANCE TRAINING (vertical loading protocol)					
Dumbbell Chest Press	3	10	Medium	0	Superset
Standing Cable Chest Press		10	Slow	60 sec	
Machine Cable Row	3	10	Medium	0	Superset
Single-Leg Cobras		10	Slow	60 sec	
Leg Press	3	10	Medium	0	Superset
Single-leg Squat		10	Slow	60 sec	
CLIENT'S CHOICE					
					Insert client's chosen exercise
COOL-DOWN					
SMT: Bilateral Calves, Piriformis, Thoracic Spine	1	1	Slow	0	Hold tender areas for 30 seconds
Static Stretch: Bilateral Calves, Hip Flexors, and Pecs	1	1	Slow	0	Hold each stretch for 30 seconds

Phase 2. Strength Endurance Training (program 3)

EXERCISE	SETS	REPS	TEMPO	REST	NOTES
WARM-UP					
SMT: Bilateral Calves, Adductors, Thoracic Spine	1	1	Slow	0	Hold tender areas for 30 seconds
Active Stretch: Bilateral Calves, Hip Flexors, and Pecs	1	8	Medium	0	Hold each stretch for 1-2 seconds
ACTIVATION (core & balance)					
Cable Chop	3	10 each	Medium	0	Keep glutes and abs engaged
Turning Step-Up to Balance (transverse plane)	3	10 each	Medium	60 sec	Maintain knee alignment
SKILL DEVELOPMENT (plyometric & SAQ)					
RESISTANCE TRAINING (vertical loading protocol)					
Dumbbell Alternating-Arm Chest Press	3	10	Medium	0	Superset
Suspended Push-up		10	Slow	60 sec	
Alternating-Arm Cable Row	3	10	Medium	0	Superset
Suspended Inverted Row		10	Slow	60 sec	
Barbell Romanian Deadlift	3	10	Medium	0	Superset
Stability Ball Hamstring Curls		10	Slow	60 sec	
CLIENT'S CHOICE					
					Insert client's chosen exercise
COOL-DOWN					
SMT: Bilateral Calves, Adductors, Thoracic Spine	1	1	Slow	0	Hold tender areas for 30 seconds
Static Stretch: Bilateral Calves, Hip Flexors, and Pecs	1	1	Slow	0	Hold each stretch for 30 seconds

Phase 3. Muscular Development Training (program 1)

EXERCISE	SETS	REPS	TEMPO	REST	NOTES
WARM-UP					
SMT: Bilateral Calves, TFL, Thoracic Spine	1	1	Slow	0	Hold tender areas for 30 seconds
Active Stretch: Bilateral Calves, Hip Flexors, and Pecs	1	8	Medium	0	Hold each stretch for 1-2 seconds
ACTIVATION (core & balance)					
Cable Lift	2	10 each	Medium	0	Keep abs engaged
Side Step-Up to Balance (frontal plane)	2	10 each	Medium	60 sec	Maintain knee alignment
SKILL DEVELOPMENT (plyometric & SAQ)					
RESISTANCE TRAINING (horizontal loading protocol)					
Goblet Squats	3	12	Medium	90 sec	Maintain proper alignment
Dumbbell Chest Press	3	12	Medium	90 sec	Maintain proper alignment
1-Arm Dumbbell Row	3	12	Medium	90 sec	Avoid shrugging
Landmine Shoulder Press	3	12	Medium	90 sec	Maintain neutral spine
CLIENT'S CHOICE					
					Insert client's chosen exercise
COOL-DOWN					
SMT: Bilateral Calves, TFL, Thoracic Spine	1	1	Slow	0	Hold tender areas for 30 seconds
Static Stretch: Bilateral Calves, Hip Flexors, and Pecs	1	1	Slow	0	Hold each stretch for 30 seconds

Phase 3. Muscular Development Training (program 2)

EXERCISE	SETS	REPS	TEMPO	REST	NOTES
WARM-UP					
SMT: Bilateral Calves, TFL, Lats	1	1	Slow	0	Hold tender areas for 30 seconds
Active Stretch: Bilateral Calves, Hip Flexors, and Lats	1	8	Medium	0	Hold each stretch for 1-2 seconds
ACTIVATION (core & balance)					
Cable Lifts	3	10	Medium	0	Keep glutes and abs engaged
Side Lunge to Balance (frontal plane)	3	8 each	Medium	60 sec	Maintain knee alignment
SKILL DEVELOPMENT (plyometric & SAQ)					
RESISTANCE TRAINING (horizontal loading protocol)					
Hex Bar Deadlift	3	10	Medium	90 sec	Maintain proper alignment
Barbell Bench Press	3	10	Medium	90 sec	Avoid shrugging
Machine Lat Pulldown	3	10	Medium	90 sec	Maintain proper alignment
Seated Dumbbell Overhead Press	3	10	Medium	90 sec	Maintain neutral spine
CLIENT'S CHOICE					
					Insert client's chosen exercise
COOL-DOWN					
SMT: Bilateral Calves, TFL, Lats	1	1	Slow	0	Hold tender areas for 30 seconds
Static Stretch: Bilateral Calves, Hip Flexors, and Lats	1	1	Slow	0	Hold each stretch for 30 seconds

Phase 3. Muscular Development Training (program 3)

EXERCISE	SETS	REPS	TEMPO	REST	NOTES
WARM-UP					
SMT: Bilateral Calves, TFL, Thoracic Spine	1	1	Slow	0	Hold tender areas for 30 seconds
Active Stretch: Bilateral Calves, Hip Flexors, and Pecs	1	8	Medium	0	Hold each stretch for 1-2 seconds
ACTIVATION (core & balance)					
Cable Lifts	3	10	Medium	0	Do not hyperextend low-back
Turning Lunge to Balance	3	8 each	Medium	60 sec	Maintain knee alignment
SKILL DEVELOPMENT (plyometric & SAQ)					
RESISTANCE TRAINING (horizontal loading protocol)					
Barbell Romanian Deadlifts	4	8	Medium	90 sec	Maintain neutral spine
Incline Dumbbell Chest Press	4	8	Medium	90 sec	Maintain neutral spine
1-Arm Dumbbell Row	4	8	Medium	90 sec	Avoid shrugging
Seated Dumbbell Overhead Press	4	8	Medium	90 sec	Maintain neutral spine
CLIENT'S CHOICE					
					Insert client's chosen exercise
COOL-DOWN					
SMT: Bilateral Calves, TFL, Thoracic Spine	1	1	Slow	0	Hold tender areas for 30 seconds
Static Stretch: Bilateral Calves, Hip Flexors, and Pecs	1	1	Slow	0	Hold each stretch for 30 seconds

Example Power Training Programs

The third level of the OPT model, power, is designed to increase the rate of force production (or speed of muscle contraction). This form of training uses the adaptations of stabilization and strength acquired in the previous phases of training and applies them at more realistic speeds and forces that the body will encounter in everyday life and sports.

Power training is usually not a common practice in the fitness environment, but it is viable and purposeful in a properly planned golf training program. Power is defined as force multiplied by velocity ($P = F \times V$). Therefore, any increase in either force or velocity will produce an increase in power. This is accomplished by either increasing the load (or force) as in progressive strength training, or increasing the speed with which a load is moved (or velocity). The combined effect is a better rate of force production in both daily and sporting activities.

To develop optimal levels of power, individuals should train with both heavy loads (85% to 100% intensity or one to five repetition maximum [RM]) and light loads (30% to 45% intensity or 8 to 10 RM) at high speeds. The focus of power training is to increase the rate of force production by increasing the number of motor units activated, the synchrony between them, and the speed at which they are excited. A client in this category will generally stay in this phase of training for a 2- to 6-week duration before cycling back through earlier phases, but this is highly dependent on the client's training goals or competitive athletic schedule.

The OPT model's power level consists of two training phases: Phase 5. Power Training and Phase 6. Maximal Power Training.

Power training can significantly impact strength and performance for golfers qualified for this type of exercise. These programs should be used with caution, but they are beneficial for golfers who have successfully completed previous phases, and modifications should be made to fit the needs and abilities of the client.

Phase 5. Power Training (program 1)

EXERCISE	SETS	REPS	TEMPO	REST	NOTES
WARM-UP					
SMT: Calves, Adductor, Thoracic Spine	1	1	Slow	0	Hold tender areas for 30 seconds each
Dynamic Stretches: Lunge with Rotation, Lateral Tube Walks	1	12	Controlled	60 sec	Maintain proper alignment
ACTIVATION (core & balance)					
Single-Leg Hop with Stabilization -front to back	2	10 each	Controlled	60 sec	Hold landing for 3-5 seconds
SKILL DEVELOPMENT (plyometric & SAQ)					
RESISTANCE TRAINING (vertical loading protocol)					
Barbell Bench Press Medicine Ball Rotation Chest Pass	4	5 10	Explosive Explosive	0 2 min	Superset
Machine Cable Row Banded Speed Row	4	5 10	Explosive Explosive	0 2 min	Superset
Barbell Squat Squat Jump	4	5 10	Explosive Explosive	0 2 min	Superset
CLIENT'S CHOICE					
					Insert client's chosen exercise
COOL-DOWN					
SMT: Calves, Adductors, Thoracic Spin	1	1	Slow	0	Hold tender areas for 30 seconds
Static Stretch: Calves, Standing Hip Flexors, Lats	1	1	Slow	0	Hold each stretch for 30 seconds

Phase 5. Power Training (program 2)

EXERCISE	SETS	REPS	TEMPO	REST	NOTES
WARM-UP					
SMT: Calves, Adductors, Lats	1	1	Slow	0	Hold tender areas for 30 seconds
Dynamic Stretches: Push-ups with Rotation, Leg Swings, Lunge with Rotation	1	10	Controlled	60 sec	Perform in a circuit
ACTIVATION (core & balance)					
SKILL DEVELOPMENT (plyometric & SAQ)					
RESISTANCE TRAINING (vertical loading protocol)					
Incline Barbell Bench Press Medicine Ball Rotation Chest Pass	4	5 10	Explosive Explosive	0 2 min	Superset
Lat Pulldown Medicine Ball Soccer Throw	4	5 10	Explosive Explosive	0 2 min	Superset
Dumbbell Overhead Press Medicine Ball Overhead Toss	4	5 10	Explosive Explosive	0 2 min	Superset
Barbell Deadlift Depth Jump	4	5 10	Explosive Explosive	0 2 min	Superset
CLIENT'S CHOICE					
					Insert client's chosen exercise
COOL-DOWN					
SMT: Calves, Adductors, Thoracic Spine	1	1	Slow	0	Hold tender areas for 30 seconds
Static Stretch: Bilateral Calves, Hip Flexors and Lats	1	1	Slow	0	Hold each stretch 30 seconds each

Phase 5. Power Training (program 3)

EXERCISE	SETS	REPS	TEMPO	REST	NOTES
WARM-UP					
SMT: Calves, Adductors, Thoracic Spine	1	1	Slow	0	Hold tender areas for 30 seconds
Dynamic Stretches: Walking Lunges with Rotation, Push-ups with Rotation, Leg Swings	1	10	Controlled	0	Perform in a circuit
ACTIVATION (core & balance)					
SKILL DEVELOPMENT (plyometric & SAQ)					
RESISTANCE TRAINING (vertical loading protocol)					
Barbell Bench Press Plyometric Push-up	5	5 10	Explosive Explosive	0 2 min	Superset
Bent-over Barbell Row Medicine Ball Soccer Throw	5	5 10	Explosive Explosive	0 2 min	Superset
Barbell Overhead Press Medicine Ball Overhead Toss	5	5 10	Explosive Explosive	0 2 min	Superset
Hack Squat Squat Jump	5	5 10	Explosive Explosive	0 2 min	Superset
CLIENT'S CHOICE					
COOL-DOWN					
SMT: Calves, Adductors, Thoracic Spine (if applicable)	1		Slow		Hold tender areas for 30 seconds
Static Stretch: Calves, Hip Flexors and Lats	1		Slow		Hold each stretch 30 seconds each

Phase 6. Maximal Power Training (program 1)

EXERCISE	SETS	REPS	TEMPO	REST	NOTES
WARM-UP					
SMT: Calves, Adductor, Thoracic Spine	1	1	Slow	0	Hold tender areas for 30 seconds each
Dynamic Stretches: Lunge with Rotation, Lateral Tube Walks	1	12	Controlled	60 sec	Maintain proper alignment
ACTIVATION (core & balance)					
SKILL DEVELOPMENT (plyometric & SAQ)					
RESISTANCE TRAINING (vertical loading protocol)					
Power Clean	4	10	Explosive	2 min	Maintain 5 kinetic chain checkpoints
Medicine Ball Rotation Chest Pass	4	10	Explosive	2 min	Finish in triple extension
Medicine Ball Soccer Throw	4	10	Explosive	2 min	Maintain a neutral spine
Medicine Ball Scoop Toss	4	10	Explosive	2 min	Keep core engaged
CLIENT'S CHOICE					
					Insert client's chosen exercise
COOL-DOWN					
SMT: Calves, Adductors, Thoracic Spin	1	1	Slow	0	Hold tender areas for 30 seconds
Static Stretch: Calves, Standing Hip Flexors, Lats	1	1	Slow	0	Hold each stretch for 30 seconds each

Phase 6. Maximal Power Training (program 2)

EXERCISE	SETS	REPS	TEMPO	REST	NOTES
WARM-UP					
SMT: Calves, Adductor, Thoracic Spine	1	1	Slow	0	Hold tender areas for 30 seconds each
Dynamic Stretches: Lunge with Rotation, Lateral Tube Walks	1	12	Controlled	60 sec	Maintain proper alignment
ACTIVATION (core & balance)					
SKILL DEVELOPMENT (plyometric & SAQ)					
RESISTANCE TRAINING (vertical training protocol)					
Push Jerk	5	10	Explosive	2 min	Maintain 5 kinetic chain checkpoints
Medicine Ball Rotation Chest Pass	5	10	Explosive	2 min	Finish in triple extension
Medicine Ball Soccer Throw	5	10	Explosive	2 min	Keep core engaged
Ice Skaters (aka Skater jumps)	5	10	Explosive	2 min	Maintain 5 kinetic chain checkpoints
CLIENT'S CHOICE					
					Insert client's chosen exercise
COOL-DOWN					
SMT: Calves, Adductors, Thoracic Spin	1	1	Slow	0	Hold tender areas for 30 seconds
Static Stretch: Calves, Standing Hip Flexors, Lats	1	1	Slow	0	Hold each stretch for 30 seconds each

Phase 6. Maximal Power Training (program 3)

EXERCISE	SETS	REPS	TEMPO	REST	NOTES
WARM-UP					
SMT: Calves, Adductor, Thoracic Spine	1	1	Slow	0	Hold tender areas for 30 seconds each
Dynamic Stretches: Lunge with Rotation, Lateral Tube Walks	1	12	Controlled	60 sec	Maintain proper alignment
ACTIVATION (core & balance)					
SKILL DEVELOPMENT (plyometric & SAQ)					
RESISTANCE TRAINING (vertical training protocol)					
Push Press	5	10	Explosive	2 min	Maintain 5 kinetic chain checkpoints
Medicine Ball Rotation Chest Pass	5	10	Explosive	2 min	Finish in triple extension
Medicine Ball Scoop Toss	5	10	Explosive	2 min	Keep core engaged
Depth Jumps	5	10	Explosive	2 min	Maintain 5 kinetic chain checkpoints
CLIENT'S CHOICE					
					Insert client's chosen exercise
COOL-DOWN					
SMT: Calves, Adductors, Thoracic Spin	1	1	Slow	0	Hold tender areas for 30 seconds
Static Stretch: Calves, Standing Hip Flexors, Lats	1	1	Slow	0	Hold each stretch for 30 seconds each